A quaternion phase-correlation counterexample

Negative answer to Farhangi–Tucker-Drob Question 3.7

Automated mathematics run `fa_banach_001`, lane 9

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Status. Candidate counterexample, likely valid; independent human review is required. The proof is exact and self-contained. A bounded search found no prior answer to the source question, but does not establish publication-level novelty.

Abstract

We give a normalized positive-definite function on the countably infinite amenable group $Q_8 \times \mathbb{Z}$ which cannot be represented as the correlation function of any circle-valued observable in a measure-preserving system. The obstruction is a rank-two extreme point of the 4×4 complex correlation elliptope obtained from four quaternion translates of a vector in $\mathbb{C}^2$. Thus property (i) in Question 3.7 of Farhangi and Tucker-Drob already fails; property (ii) cannot repair the representation.

1 Source question

The source is S. Farhangi and R. Tucker-Drob, *Asymptotic dynamics on amenable groups and van der Corput sets*, arXiv:2409.00806v3 [1]. The question appears as Question 3.7 on PDF page 18 (the source label is `MainConjecture`). The preceding paragraph says that a positive answer for amenable groups would complete an equivalence in their Theorem 3.5.

> Furthermore, in Theorem 3.5, we would like to show that (i)-(iv) are equivalent for any amenable group. This would follow from our proof provided the following questions has a positive answer for all amenable G.
>
> **Question 3.7.** Let G be a countable group and let $\phi : G \to \mathbb{C}$ be a positive definite sequence for which $\phi(e) = 1$. Does there exists a measure preserving system $(X, \mathscr{B}, \mu, (\tau_g)_{g\in G})$ and a measurable $f : X \to \mathbb{S}^1$ for which the following holds:
>
> (i) $\phi(h) = \langle \tau_h f, f\rangle$ for all $h \in G$.
>
> (ii) $\int_X f d\mu = 0$ if and only if f is orthogonal to the subspace of $L^2(X, \mu)$ of T-invariant functions.

Figure 1: Question 3.7 and its lead-in, rendered from page 18 of the source PDF.

In transcription, the question asks: given a countable group G and a positive-definite $\phi : G \to \mathbb{C}$ with $\phi(e) = 1$, must there exist an m.p.s. $(X, \mathcal{B}, \mu, (\tau_g)_{g\in G})$ and measurable $f : X \to \mathbb{S}^1$ such that

(i) $\phi(h) = \langle \tau_h f, f\rangle$ for every $h \in G$;

(ii) $\int_X f\, d\mu = 0$ if and only if f is orthogonal to the invariant subspace of $L^2(X, \mu)$?

Throughout, as in the source, Hilbert-space inner products are linear in the first variable. Thus positive definiteness means

$$\sum_{i,j} c_i \overline{c_j} \phi(g_j^{-1} g_i) \geq 0.$$

2 Proof intuition

Four unit vectors have a correlation matrix C. If the vectors are four translates of a circle-valued random variable, then pointwise they are four complex phases; consequently C is an average of rank-one phase matrices. We construct four quaternion translates in $\mathbb{C}^2$ whose matrix C has rank two but is extreme among *all* complex correlation matrices. An extreme matrix cannot be a nontrivial average of other correlation matrices. It therefore cannot be an average of rank-one phase matrices. The quaternion coefficient function realizes precisely these four entries, forcing the contradiction.

3 An extremality lemma

Let

$$\mathcal{E}_n = \{C \in M_n(\mathbb{C}) : C = C^*,\ C \geq 0,\ C_{ii} = 1\}$$

be the complex correlation elliptope. The following standard criterion is included with its proof; see Li and Tam [2] for background.

Lemma 1 (Projector-spanning criterion)**.** *Let $v_1, \ldots, v_n$ be unit vectors spanning a finite-dimensional complex Hilbert space H, and let $C_{ij} = \langle v_i, v_j \rangle$. If the rank-one orthogonal projectors $P_i = v_i v_i^*$ span* $\mathrm{Herm}(H)$ *over $\mathbb{R}$, then C is an extreme point of $\mathcal{E}_n$.*

Proof. Suppose $C = (A+B)/2$ with $A, B \in \mathcal{E}_n$. Since $0 \leq A, B \leq 2C$, the usual dominated Gram-factorization gives positive operators S, T on H such that

$$A_{ij} = \langle S v_i, v_j \rangle, \qquad B_{ij} = \langle T v_i, v_j \rangle.$$

For completeness, this factorization is obtained by defining the quadratic form associated to A on linear combinations of the v_i; domination by the Gram form of C makes it well-defined and bounded, hence represented by a positive operator. The same applies to B.

The equality $(A+B)/2 = C$ and the spanning assumption on the vectors imply $(S+T)/2 = I_H$. Since every diagonal entry of A equals one,

$$0 = A_{ii} - C_{ii} = \langle (S - I_H) v_i, v_i \rangle = \mathrm{tr}((S - I_H) P_i) \quad (1 \leq i \leq n).$$

The Hermitian operator $S - I_H$ is therefore Hilbert–Schmidt orthogonal, over $\mathbb{R}$, to a spanning set of $\mathrm{Herm}(H)$. Hence $S = I_H$. It follows that $T = I_H$ and $A = B = C$, proving extremality. $\square$

4 The counterexample

Let

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

be the Pauli matrices, and put

$$Q_8 = \{\pm I,\ \pm iX,\ \pm iY,\ \pm iZ\} \subset U(2).$$

This is the quaternion group. Set $G = Q_8 \times \mathbb{Z}$, a countably infinite amenable group, and let $\pi(q,m) = q$ be its two-dimensional unitary representation.

Set $r = 1/\sqrt{3}$ and choose

$$\xi = \begin{pmatrix} \sqrt{(1+r)/2} \\ e^{i\pi/4}\sqrt{(1-r)/2} \end{pmatrix}.$$

Then $\|\xi\| = 1$ and its three Pauli expectations are all r:

$$\langle X\xi,\xi\rangle = \langle Y\xi,\xi\rangle = \langle Z\xi,\xi\rangle = r.$$

Define

$$\phi(g) = \langle \pi(g)\xi,\xi\rangle \qquad (g \in G).$$

As a coefficient of a unitary representation, ϕ is positive definite in the source convention, and $\phi(e) = 1$.

Theorem 1. *There is no measure-preserving G-system and no measurable $f : X \to \mathbb{S}^1$ satisfying*

$$\phi(h) = \langle \tau_h f, f\rangle \qquad (h \in G).$$

Consequently Question 3.7 has a negative answer, even for a countably infinite amenable group.

Proof. Let

$$q_1 = I, \quad q_2 = iX, \quad q_3 = iY, \quad q_4 = iZ, \qquad g_j = (q_j, 0) \in G,$$

and set $v_j = q_j\xi$. If $P_j = v_j v_j^*$, then the coordinates of $2P_j$ in the ordered real basis (I, X, Y, Z) are the rows of

$$M = \begin{pmatrix} 1 & r & r & r \\ 1 & r & -r & -r \\ 1 & -r & r & -r \\ 1 & -r & -r & r \end{pmatrix}.$$

Indeed, conjugation by iX, iY, or iZ fixes the corresponding Bloch coordinate and flips the other two. Direct calculation gives

$$\det M = -\frac{16}{3\sqrt{3}} \neq 0.$$

Thus $P_1, \ldots, P_4$ span $\mathrm{Herm}(2)$. By Lemma 1, their Gram matrix C is extreme in $\mathcal{E}_4$. Explicitly,

$$C = (\langle v_i, v_j\rangle)_{i,j=1}^4 = \begin{pmatrix} 1 & -ir & -ir & -ir \\ ir & 1 & -ir & ir \\ ir & ir & 1 & -ir \\ ir & -ir & ir & 1 \end{pmatrix}.$$

The vectors span $\mathbb{C}^2$, so $\operatorname{rank} C = 2$; equivalently its spectrum is $\{2, 2, 0, 0\}$.

Assume, for contradiction, that the requested circle-valued realization exists. Put $z_j = \tau_{g_j} f$. Since the Koopman operators form a unitary representation and the source convention is linear in the first variable,

$$\langle z_i, z_j\rangle = \langle \tau_{g_j^{-1}g_i} f, f\rangle = \phi(g_j^{-1}g_i) = \langle q_i\xi, q_j\xi\rangle = C_{ij}.$$

For almost every x, all four coordinates of $z(x) = (z_1(x), \ldots, z_4(x))$ lie on $\mathbb{T}$. Hence

$$C = \int_X z(x)z(x)^* \, d\mu(x).$$

Every integrand is a rank-one member of $\mathcal{E}_4$. Because the phase matrices zz^* form a compact set and $\mathcal{E}_4$ is finite-dimensional, the integral belongs to their convex hull. Extremality of C in the larger set $\mathcal{E}_4$ forces every nonzero term in any finite convex representation to equal C. This is impossible: each phase matrix has rank one, whereas C has rank two. The contradiction proves the theorem. $\square$

Why condition (ii) is irrelevant. The construction disproves the correlation representation demanded in (i), before any condition involving invariant functions or the integral of f is considered.

5 Verification and limitations

The companion script `code/verify_quaternion_elliptope.py` checks all 64 products in the matrix model of Q_8, the normalization and Bloch coordinates of ξ, the projector-coordinate determinant and rank, the displayed Gram matrix, and its spectrum. It returns projector-span rank 4 and Gram rank 2. These floating-point checks are safeguards against transcription and sign errors; the determinant and rank arguments in the proof are exact and do not depend on the script.

The construction uses a representation of $Q_8 \times \mathbb{Z}$ that is trivial on the $\mathbb{Z}$ factor. This is allowed: the question imposes no faithfulness or mixing assumption. The example therefore addresses both the literal "countable group" statement and the countably infinite amenable setting motivating it.

6 Bounded novelty check

On 19 July 2026, the run registry and local full-text indexes were searched for arXiv:2409.00806 and the core terms *circle-valued*, *positive definite*, *phase correlation*, *quaternion*, and *extreme correlation*. Web/arXiv searches used the exact title and question phrase, the authors together with "Question 3.7", and close variants. Searches also looked for later work explicitly answering the question. They found the standard extreme-correlation background [2], but no explicit answer to Question 3.7 and no application of this obstruction to it.

This search was bounded rather than exhaustive. Accordingly, the packet claims a complete mathematical counterexample subject to human verification, but makes no claim of publication-level novelty.

7 Human-review recommendation

The first review priority is the identity $C_{ij} = \phi(g_j^{-1} g_i)$ and its Koopman counterpart, because it is where left/right and inner-product conventions enter. The second is the extreme-point step converting a phase-valued realization into a forbidden rank-one barycenter. If those two short steps pass independent review, the argument is a full negative answer to the question as stated.

References

[1] S. Farhangi and R. Tucker-Drob, *Asymptotic dynamics on amenable groups and van der Corput sets*, arXiv:2409.00806v3, 2025; Groups, Geometry, and Dynamics (2026), https://arxiv.org/abs/2409.00806.

[2] C.-K. Li and B.-S. Tam, A note on extreme correlation matrices, *SIAM J. Matrix Anal. Appl.* **15** (1994), no. 3, 903–908, https://doi.org/10.1137/S0895479892240683.